

UMMAY MAIMONA CHAMAN

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OBJECTIVE

Computer Science undergraduate with hands-on experience in full-stack web development (MERN, Django, React), UI/UX design, and AI-powered tools including custom GPT integration and RAG systems. Passionate about building practical digital solutions across web, mobile, and AI platforms. Eager to contribute to real-world projects, bringing strong development skills, a fast-learning mindset, and a portfolio of deployed applications.

EDUCATION

BRAC University

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

2022 – 2026 (Expected)

- **CGPA:** Ongoing
- **Relevant Coursework:** Machine Learning, Deep Learning, Neural Network, Computer Vision, Data Science, Algorithms and Data Structures, Microprocessors, Robotics, Web Development
- **Undergraduate Thesis:** *A twin-aware multimodal deep learning framework with optimized late fusion for early prediction of adolescent anxiety disorder. Under review at PLOS ONE.* Available as a preprint: medRxiv
- **Supervisor:** Dr. Md. Golam Rabiul Alam **Co-Supervisor:** Mr. Rafeed Rahman

Holy Cross College

Dhaka, Bangladesh

HSC, Science

2019 – 2021

- **GPA:** 5.00 / 5.00

Monipur High School and College

Dhaka, Bangladesh

SSC, Science

2017 – 2019

- **GPA:** 5.00 / 5.00

EXPERIENCE

Undergraduate Researcher

Jan 2024 – 2026

BIOSE, BRAC University

Dhaka, Bangladesh

- Conducting AI-driven biomedical research in medical imaging, bioinformatics, and disease prediction
- Utilizing tools such as IEEE Xplore, ACM DL, PubMed, Mendeley, and Zotero for literature analysis

Director

Sep 2023 – Feb 2026

BURed Club (Research & Development)

- Led undergraduate research initiatives and mentored students in academic writing and publication
- Organized workshops, seminars, and interdisciplinary research collaborations

TECHNICAL PROJECTS

DataFlow Studio | Django, React, TypeScript, Pandas, REST API, SQL, Docker | [GitHub](#) | [Live Demo](#) **2026**

- Built a full-stack visual ETL pipeline builder using **Django** and **React + TypeScript**, enabling drag-and-drop construction of complex data transformation pipelines.
- Implemented an advanced transformation suite (Filter, Formula, Join, GroupBy, Cleaning) powered by **Pandas**, with live data preview at every pipeline node.

AgriNetwork | React, Node.js, Express, MongoDB, REST API, JWT, Vercel | [GitHub](#) | [Live Demo](#) **2026**

- Developed a full-stack MERN agricultural networking platform with a **React + Vite** frontend and **Express/MongoDB** backend supporting community interactions.
- Designed RESTful APIs, documented endpoints using Postman, integrated GitHub Actions CI/CD, and deployed the application on Vercel.

ApneaSense | Python, XGBoost, SHAP, Machine Learning | [GitHub](#) **2026**

- Developed an ensemble machine learning framework for early sleep apnea risk prediction using **XGBoost** and feature engineering.
- Integrated **SHAP** explainability to provide clinically interpretable feature importance and prediction insights.
- Improved diagnostic interpretability while maintaining strong predictive performance.

FaceGuard <i>PyTorch, Deep Learning, MesoNet, Capsule Networks</i> GitHub – Built a multimodal deepfake detection system combining CNN-based MesoNet and Capsule Networks. – Implemented an automated forensic reporting pipeline for media authenticity verification. – Enhanced robustness against manipulated facial media through hybrid model fusion.	2026
CNN Image Classifier <i>TensorFlow.js, Deep Learning, JavaScript</i> GitHub Live Demo – Developed a browser-based deep learning workbench using TensorFlow.js for real-time image classification. – Implemented a CNN training pipeline with GradCAM explainability visualization. – Enabled privacy-first client-side inference, eliminating server dependency.	2025
SkillPocket <i>MERN Stack, React, Node.js, MongoDB</i> GitHub Live Demo – Architected a full-stack skill exchange platform with a React frontend and Node.js backend. – Implemented authentication, scheduling, and feedback systems for user interaction. – Designed RESTful APIs with MongoDB for scalable data management.	2025
Unsupervised Neural Networks <i>Python, SOM, Autoencoders</i> GitHub – Implemented Self-Organizing Maps and Autoencoders for pattern discovery in clinical datasets. – Achieved a 15% improvement over baseline clustering methods. – Applied unsupervised feature learning for efficient data representation.	2025
CurveMaker <i>Flask, Python, Matplotlib, SciPy</i> GitHub Live Demo – Created a chart-building web application with real-time previews and multiple curve types. – Implemented data validation and interpolation using SciPy. – Enabled PNG/SVG export with high-quality rendering.	2024
Multilingual RAG Chatbot: Document Intelligence System <i>Python, FAISS, OCR</i> GitHub Live Demo – Built a multilingual (Bengali/Banglish & English) RAG system for contextual question answering over PDFs and images using OCR and semantic search. – Integrated Tesseract OCR, FAISS, and the local LLM <code>microsoft/phi-2</code> for privacy-preserving document intelligence. – Developed a FastAPI backend with a Streamlit frontend and deployed the application using Docker on Hugging Face Spaces.	2026
RxCare Pharmacy Management <i>PHP, MySQL, HTML/CSS/JS</i> GitHub – Designed a full-featured pharmacy management system. – Built a dynamic frontend integrated with the backend. – Implemented role-based access control and automated alerts.	2024
Hardware Projects	
Digital Judging Platform <i>Microcontroller, Embedded C, STM32</i> GitHub – Designed a role-based voting system with weighted scoring and real-time result calculation. – Implemented validation and automated winner determination logic.	2024
Smart Plant Watering System <i>Arduino, Sensors, Robotics</i> – Built an autonomous irrigation system with environmental sensing and adaptive watering. – Integrated multi-sensor inputs for intelligent control decisions. – Implemented navigation and obstacle avoidance logic.	2025
Autonomous Restroom Management <i>Arduino, IoT, Telegram API</i> – Built an Arduino-based restroom monitoring system for leakage detection and odor management. – Integrated the Telegram API for real-time maintenance alerts. – Improved facility monitoring through an automated alert system.	2025

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, C++, R, SQL, PHP

Machine Learning & AI: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, Matplotlib

Deep Learning Specialization: CNN, RNN, Anomaly Detection (PatchCore), FL, GradCAM, CV

Web Development: MERN Stack (MongoDB, Express.js, React, Node.js), REST API, TensorFlow.js

Databases: MongoDB, MySQL, MariaDB

Robotics & Embedded Systems: Arduino, Raspberry Pi, STM32, Sensors, Microcontrollers, LTSpice, PSpice

Development Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, LaTeX, Linux

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy

Design & Documentation: Adobe Photoshop, Illustrator, Canva, Microsoft Office Suite, Technical Writing

PROFESSIONAL TRAINING & CERTIFICATIONS

- **Machine Learning Fundamentals** — Simplilearn (Advanced ML code and Theory).
- **Introduction to Artificial Intelligence** — University of Helsinki (Comprehensive AI course).
- **Python Programming** — Kaggle, HackerRank (Advanced programming certifications).
- **JavaScript Foundations** — Cisco Networking Academy (Web development certification).
- **Cybersecurity Fundamentals** — IBM P-Tech (Security).
- **Prompt Engineering Basics** — Intellectium (AI interaction and optimization).
- **Digital Marketing Essentials** — Google Digital Garage (Marketing analytics).
- **HTML & Web Design** — BUP IEEE Women in Engineering (Frontend development).

PUBLICATIONS & MANUSCRIPTS

- Noor, A., **Chaman, U. M.**, & Kabir, M. S. Federated Learning in Healthcare: A Comprehensive Survey on Privacy, Scalability, and Clinical Applications. *ICT Express (Q1 Journal)*. [Paper] [Accepted].
- Kabir, M. S., Noor, A., **Chaman, U. M.**, & Mehreen, N. Robust Multi-Backbone Hybrid Fusion for Chest X-Ray Pneumonia Detection. *QPAIN 2026 Conference*. [Paper] [Accepted].

HONORS & AWARDS

- **Bronze Award** — Duke of Edinburgh International Award Program (2023). Certificate
- **Bronze Award** — The Queen's Commonwealth Essay Competition, Senior Category (2021). Certificate
- **Pre-finalist** — International Astronomy and Astrophysics Competition (IAAC) (2022). Certificate
- **Pre-finalist** — International Youth Math Challenge (IYMC) (2021). Certificate
- **Silver Badge** — HackerRank